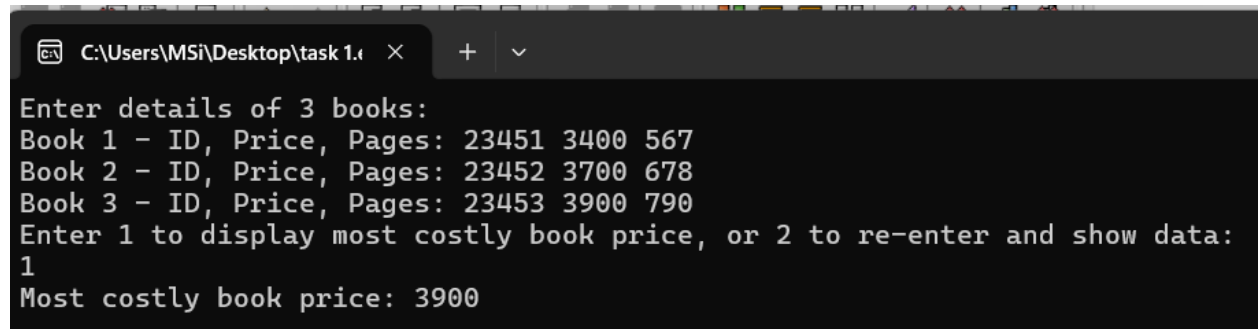


REPORT DOCUMENT

- TASK 1:

```
task 1.cpp
1  #include <iostream>
2  using namespace std;
3  struct Book {
4      int bookID;
5      float price;
6      int pages;
7  };
8  int main() {
9      Book b1, b2, b3;
10     Book *ptr;
11     cout << "Enter details of 3 books:\n";
12     cout << "Book 1 - ID, Price, Pages: ";
13     cin >> b1.bookID >> b1.price >> b1.pages;
14     cout << "Book 2 - ID, Price, Pages: ";
15     cin >> b2.bookID >> b2.price >> b2.pages;
16     cout << "Book 3 - ID, Price, Pages: ";
17     cin >> b3.bookID >> b3.price >> b3.pages;
18     int choice;
19     cout << "Enter 1 to display most costly book price, or 2 to re-enter and show data: "<<endl;
20     cin >> choice;
21     if (choice == 1) {
22         ptr = &b1;
23         if (b2.price > ptr->price) ptr = &b2;
24         if (b3.price > ptr->price) ptr = &b3;
25         cout << "Most costly book price: " << ptr->price << endl;
26     }
27     else if (choice == 2) {
28         Book bNew;
29         ptr = &bNew;
30         cout << "Enter new book details (Book ID, Price, Pages): ";
31         cin >> ptr->bookID >> ptr->price >> ptr->pages;
32         cout << "--- New Book Record ---"<<endl;
    }
```



```
Enter details of 3 books:
Book 1 - ID, Price, Pages: 23451 3400 567
Book 2 - ID, Price, Pages: 23452 3700 678
Book 3 - ID, Price, Pages: 23453 3900 790
Enter 1 to display most costly book price, or 2 to re-enter and show data:
1
Most costly book price: 3900
```

- TASK 2:

task 2 new.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Citizen {
4      char cnicID[20];
5      char name[30];
6      char city[20];
7      int age;
8  };
9  int main() {
10     Citizen c;
11     Citizen *ptr;
12     ptr = &c;
13     cout << "Enter CNIC ID: ";
14     cin >> ptr->cnicID;
15     cout << "Enter Name: ";
16     cin >> ptr->name;
17     cout << "Enter City: ";
18     cin >> ptr->city;
19     cout << "Enter Age: ";
20     cin >> ptr->age;
21     int choice;
22     cout << "Press 1 to update city of residence, 0 to display record: ";
23     cin >> choice;
24     if (choice == 1) {
25         cout << "Enter new city: ";
26         cin >> ptr->city;
27         cout << "--- Updated NADRA Record ---" << endl;
28     } else {
29         cout << "--- NADRA Record ---" << endl;
30     }
31     cout << "CNIC ID : " << ptr->cnicID << endl;
32     cout << "Name      : " << ptr->name << endl;
```

```
16     cout << "Enter CNIC ID: ";

Enter CNIC ID: 6788
Enter Name: tawasal
Enter City: lahore
Enter Age: 18
Press 1 to update city of residence, 0 to display record: 0

--- NADRA Record ---
CNIC ID : 6788
Name      : tawasal
City      : lahore
Age       : 18
```

- TASK 3:

```
task 3.cpp
1  #include<iostream>
2  using namespace std;
3  struct PetrolPump {
4      char name[20];
5      float pricePerLiter;
6      int availableLiters;
7  };
8  void input(PetrolPump *p) {
9      cout << "Enter Pump Name: ";
10     cin >> p->name;
11     cout << "Enter Price per Liter: ";
12     cin >> p->pricePerLiter;
13     cout << "Enter Available Liters: ";
14     cin >> p->availableLiters;
15 }
16 void display(PetrolPump *p) {
17     cout << "Pump Name: " << p->name << endl;
18     cout << "Price: " << p->pricePerLiter << endl;
19     cout << "Stock: " << p->availableLiters << " liters" << endl;
20 }
21 int main() {
22     PetrolPump p;
23     input(&p);
24     display(&p);
25     return 0;
26 }
```

C:\Users\MSi\Desktop\task 3. × + ▾

```
Enter Pump Name: Atoock
Enter Price per Liter: 500
Enter Available Liters: 70
Pump Name: Atoock
Price: 500
Stock: 70 liters
```

- CODE 4:

task 4.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Bill {
4      int accountNo;
5      char ownerName[30];
6      int unitsConsumed;
7      float ratePerUnit;
8      float totalBill;
9  };
10 void inputBill(Bill *b) {
11     cout << "Enter Account No : ";
12     cin >> b->accountNo;
13     cout << "Enter Owner Name : ";
14     cin >> b->ownerName;
15     cout << "Units Consumed : ";
16     cin >> b->unitsConsumed;
17     cout << "Rate Per Unit (Rs): ";
18     cin >> b->ratePerUnit;
19 }
20 void calculateBill(Bill *b) {
21     b->totalBill = b->unitsConsumed * b->ratePerUnit;
22     if (b->unitsConsumed > 300) {
23         b->totalBill += b->totalBill * 0.15;
24     }
25 }
26 void displayBill(Bill *b) {
27     float baseAmount = b->unitsConsumed * b->ratePerUnit;
28     cout << "\n===== LESCO Bill =====" << endl;
29     cout << "Account No : " << b->accountNo << endl;
30     cout << "Owner Name : " << b->ownerName << endl;
31     cout << "Units Used : " << b->unitsConsumed << endl;
32     cout << "Rate/Unit : Rs. " << b->ratePerUnit << endl;
```

C:\Users\MSi\Desktop\task 4. × + ∨

```
=== LESCO Billing System ===
Enter Account No : 567
Enter Owner Name : tawasal
Units Consumed : 350
Rate Per Unit (Rs): 25

===== LESCO Bill =====
Account No : 567
Owner Name : tawasal
Units Used : 350
Rate/Unit : Rs. 25
Base Amount : Rs. 8750
Surcharge 15%: Rs. 1312.5
Total Bill : Rs. 10062.5
```

- HOME TASK 1:

```
home task 1.cpp
1  #include<iostream>
2  using namespace std;
3  struct Patient {
4      int patientID;
5      char name[30];
6      float temperature;
7      int severity;
8  };
9  Patient* mostCritical(Patient arr[], int n) {
10     Patient *critical = &arr[0];
11     bool tie = false;
12     for (int i = 1; i < n; i++) {
13         if (arr[i].severity > critical->severity) {
14             critical = &arr[i];
15             tie = false;
16         } else if (arr[i].severity == critical->severity) {
17             tie = true;
18         }
19     }
20     if (tie) {
21         cout << " ?? ALERT: Multiple Critical Patients!" << endl;
22         for (int i = 0; i < n; i++) {
23             if (arr[i].severity == critical->severity) {
24                 cout << "ID : " << arr[i].patientID << endl;
25                 cout << "Name : " << arr[i].name << endl;
26                 cout << "Temp : " << arr[i].temperature << " C" << endl;
27                 cout << "Severity : " << arr[i].severity << "/10" << endl;
28                 cout << endl;
29             }
30         }
31         return critical;
32     }
```

OUTPUTS:

```
C:\Users\MSi\Desktop\home  X + v
=== Lahore General Hospital - Emergency Triage ===
Enter number of patients: 3
Patient 1:
ID: 567
Name: Ahmed
Temp: 78
Severity: 7
Patient 2:
ID: 568
Name: Ali
Temp: 89
Severity: 8
Patient 3:
ID: 569
Name: Bilal
Temp: 90
Severity: 9

--- Most Critical Patient ---
?? ALERT: Immediate Attention Required!
ID      : 569
Name    : Bilal
Temp    : 90 C
Severity : 9/10

Enter Patient ID to update (0 to stop): 568
Enter new temperature: 93
Enter new severity    : 9

--- Updated Record ---
```

--- Updated Record ---

ID : 568
Name : Ali
Temp : 93 C
Severity : 9/10

--- New Most Critical Patient ---

ALERT: Multiple Critical Patients!

ID : 568
Name : Ali
Temp : 93 C
Severity : 9/10

ID : 569
Name : Bilal
Temp : 90 C
Severity : 9/10

Update another patient? (y/n): n

=== Final Triage Summary ===

ID: 567	Name: Ahmed	Temp: 78 C	Severity: 7
ID: 568	Name: Ali	Temp: 93 C	Severity: 9
ID: 569	Name: Bilal	Temp: 90 C	Severity: 9
